

Unit 1 Lesson 2



Merit Education Group

Translating Verbal Phrases (Writing Expressions)

Algebra 1 · Pacing ID: WritingExpressions

Standard: 9.3.5.7

Standard: 9.3.5.8

Lesson Overview

Objectives	• Translate verbal phrases into algebraic expressions. • Translate algebraic expressions into verbal phrases.
Key Vocabulary	Phrase, Sum, Difference, Product, Quotient, Twice, Half
Skills Targeted	Phrase-to-expression translation; keyword matching; reverse translation
Materials	Whiteboards, markers, erasers (1 set per student) · Direct Instruction slide deck · Independent Worksheet (T1/T2/T3) · Prime Time Expression Race cards/tokens · Exit Ticket slips · Visual keyword anchor chart · Sentence frames handout · Bilingual vocab cards

Phase 1: Activate & Explore 16 min

Bell Ringer

4 min

Display slide with header **Whiteboards Ready!** Students solve independently on whiteboards, reviewing variable and coefficient identification from L1. Review answers collectively on teacher cue.

Companion: See U1L02_WritingExpressions_BellRinger.md.

Direct Instruction

6 min

1. Algebraic Phrase:

A mathematical phrase containing numbers, variables, and operations but no equals sign.

2. Keyword Mapping — Addition:

Sum, Increased by, More than $\rightarrow +$

3. Keyword Mapping — Subtraction:

Difference, Decreased by, Less than, Subtracted from $\rightarrow -$

4. Keyword Mapping — Multiplication:

Product, Twice, Half of, Times $\rightarrow \times$ or juxtaposition

5. Keyword Mapping — Division:

Quotient, Divided by, Ratio $\rightarrow \div$ or fraction bar

6. Critical Nuance:

“Less than” and “Subtracted from” flip the order. Example: “Five less than x ” is $x - 5$, not $5 - x$.

Precision Language: We do not simply guess keywords — we map the **structure** of the phrase. “Less than” signals order reversal: the second quantity appears first in the expression. “Twice” always signals multiplication by 2, never addition.

 Culture & Career Spotlight

6 min

Cultural Context

No culture slide this lesson.

Career: Merchandise Planner — Maurices, Duluth

Education: BS (Business/Supply Chain) |
Salary: \$74,120/year
Merchandise Planners use algebraic expressions daily to manage inventory and profitability. A planner reads: “Total cost is the sum of wholesale price per item and shipping.” If w = wholesale price and s = shipping, the expression is $w + s$. Profit margin expression: $2b - r$ (twice base rate minus return fees).

Phase 2: Practice & Apply 26 min

 Guided Practice

8 min

Example 1 (Additive/Subtractive Keywords):

Phrase: “The sum of a number and twelve.”

GEMDAS Reminder: No order conflict; direct translation.

Solution: Let n = the number. Expression: $n + 12$.

Example 2 (Multiplicative & Order Matters):

Phrase: **“Twice a number decreased by four.”**

GEMDAS Reminder: Multiplication precedes subtraction.

Solution: Let x = the number. Expression: $2x - 4$.

Check: Is it $(2x) - 4$ or $2(x - 4)$? Decreased by four follows the doubling. Result: $2x - 4$.

Example 3 (Reverse Translation):

Expression: $\frac{y}{5} + 8$

GEMDAS Reminder: Identify operations present: division and addition.

Solution: “The quotient of a number and five, increased by eight.” OR “Eight more than the quotient of y and 5.”

GEMDAS Reminder: When translating multi-step phrases, identify the **last** operation performed in the verbal description — that operation becomes the outermost structure in the expression. Multiplication and division bind tighter than addition and subtraction.

 Independent Worksheet + Circulate & Support

18 min

Tiered Structure:**T1 Foundational**

Direct keyword matching; single operation phrases. Sample: “The product of 7 and m ” $\rightarrow 7m$.

T2 Application

Multi-step phrases; reverse translation; order traps. Sample: “Five subtracted from three times a number” $\rightarrow 3n - 5$.

T3 Extension

Contextual application (Career link); complex structure. Maurices Scenario: cost = half retail price plus tax t . Write expression in terms of retail price r . $\rightarrow \frac{1}{2}r + t$.

Teacher Action: Circulate systematically. Monitor T3 students for variable definition clarity. Check T1/T2 for “less than/subtracted from” errors using redirect questions. Verify SPED/ELL students are utilizing anchor charts and sentence frames.

Phase 3: Assess & Reflect 8 min
 Game — Prime Time: Expression Race

5 min

Zero-Device Activity. Pairs receive a deck of Phrase Cards and Expression Cards. Students race to match verbal phrases to correct algebraic expressions. First pair to clear the deck wins. Teacher reveals matches; class discusses one tricky pairing (e.g., difference vs. less than).

Zero-device activity — card sets pre-laminated.

 Exit Ticket

2 min

1. Write an expression for: “**Half of a number increased by six.**”

Answer: $\frac{1}{2}n + 6$ or $0.5n + 6$.

2. Translate into words: $4m - 9$.

Acceptable: “Four times a number decreased by nine” OR “Nine less than four times m .”

Today & Tomorrow

1 min

Today: We translated verbal phrases to algebraic expressions and identified keywords like sum, product, and quotient.

Tomorrow: Lesson 3 — GEMDAS Basics: We will evaluate these expressions using order of operations. Bring your whiteboards!

SPED & ELL Cross-Reference

S

[SPED

Provide visual keyword anchor chart with color-coded operations. Use sentence frames for verbal-to-expression translation (e.g., “The expression is plus .”). Allow use of algebra tiles during Guided Practice to model coefficients.]

L

[ELL

Display bilingual vocabulary cards (sum/ suma, difference/diferencia). Pair with peer support for verbal translation tasks. Emphasize cognates where applicable; highlight false friends (e.g., “product” vs. consumer product).]

Misconceptions & Redirect Questions

	Misconception	Redirect Question
Corrective Strategy	Order Reversal: Writes $x - 5$ for “Five subtracted from a number.”	“Which quantity is the starting amount? Which is being taken away?”
Lead to $n - 5$. Use sentence frames to reverse operation order.	“Twice” as Addition: Interprets “Twice a number” as $x + 2$	“If you have twice as many apples, do you add two or double the amount?”
Lead to $2x$. Emphasize doubling vs. adding.	Variable Selection: Uses multiple variables in one expression unnecessarily	“The phrase refers to ‘a number.’ Should we use x or y?”
Consistency check — one variable per single-number phrase.		

Materials Checklist

☐ Whiteboards, markers, erasers (1 set per student)

☐ Independent Worksheet (T1 / T2 / T3 versions)

☐ Exit Ticket slips

☐ Sentence frames handout

☐ Direct Instruction slide deck (navy headings)

☐ Prime Time Expression Race cards/tokens (zero-device)

☐ Visual keyword anchor chart (color-coded operations)

☐ Bilingual vocab cards